

Task	CTC	Rungs	Mask-mixing gain	
			Mean	At deepest
BEIR FiQA	$O_T(N)$	5	+0.011	-0.001
Outlier (scale- k)	$O_T(NM)$	5	+0.505	+0.108
QDmatch (FiQA) [‡]	$O_T(N^2)$	5	+0.031	+0.006
Reordering ^{†‡}	$O_T(N^2)$	4	+0.182	+0.000
Contradiction [†]	$O_T(N^2)$	5	+0.791	+0.699

Table 1: **What curriculum mask-mixing buys, by corpus-traversal complexity.** Both chunked arms use the identical document-chunked mask at evaluation time and differ only in training: *+Mix* anneals a per-step unrestricted-attention coin $p_{\text{standard}}: 0.8 \rightarrow 0$, *Chunked* never sees an unrestricted step. *Gain* is $(+Mix) - (Chunked)$. Qwen3.5-4B, identical shards, identical ladders, matched hyperparameters; *eval_size* = 500 per rung, so a gain under ~ 0.04 is within noise. *Mean* averages the per-rung gain over the ladder; the deepest rung is 32k for every task except reordering, whose ladder ends at 16k. Mixing is worth nothing on $O_T(N)$ retrieval (FiQA, +0.011 mean) and is the difference between learning the task and not learning it at all on contradiction (+0.791) and outlier detection (+0.505). *The two quantities answer different questions and disagree where the task itself has collapsed*: reordering and QDmatch both lose their gain at depth only because every arm has hit the floor there – at 2k the same gains are +0.473 and -0.073. [†]Pure-chunked training cross-entropy never descended on these two tasks (contradiction: $1.175 \rightarrow 1.071$, flat across the whole run), while FiQA and QDmatch on the identical recipe converged normally – so the collapse is task-specific rather than an infrastructure failure, but it reflects a failure to fit the training set and is not by itself a statement about what the mask can represent. [‡]Floor-limited: the *full-attention* arm is itself below 0.15 at this rung, so neither chunked arm has headroom and the gain is uninformative.